

- 4 At the top of a cliff of height 100 m, a boy threw a rock upwards with velocity of 15 m s^{-1} . How much time later should he drop a second rock from rest so that both rocks will arrive simultaneously at the bottom of the cliff?

A 1.8 s

B 2.7 s

C 4.5 s

D 6.3 s

Ans: A

For rock dropped:

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$s_y = \frac{1}{2} a_y t^2 \text{ since } u = 0$$

$$t_1 = \sqrt{(2s_y / a_y)} = \sqrt{[2(100) / 9.81]} = 4.51 \text{ s}$$

For rock thrown upwards:

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$100 = (-15) t + \frac{1}{2} (9.81) t^2$$

$$t_2 = 6.30 \text{ s}$$

$$t_2 - t_1 = 6.30 - 4.51 = 1.8 \text{ s}$$